Week 6 Worksheet Solutions

MTH 317H — Week 6

1. Carryover: disproving injectivity

Prove or disprove: $f\colon \mathbb{R} \to \mathbb{R}$, defined by $f(x) = x^3 - x$, is injective. If you disprove it, exhibit two distinct inputs with the same output.

Solution. The claim is false: $0 \neq 1$, but $f(0) = 0 = f(1)$. Thus two distinct inputs have the same output, so f is not injective.

2. Carryover: proving surjectivity

Prove that $g\colon \mathbb{R}^2 \to \mathbb{R}^2$, defined by

$$g(x, y) = (x + y, x - y),$$

is surjective. Start with an arbitrary target $(u, v) \in \mathbb{R}^2$, find an input (x, y), and check it.

After completing the original problem: Is g also injective? If so, give its inverse and verify both compositions.

Solution. Let $(u, v) \in \mathbb{R}^2$ be arbitrary. Solving $x + y = u$ and $x - y = v$ gives $x = (u + v)/2$ and $y = (u - v)/2$. Both are real, and

$$g\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = (u, v).$$

Every target has a preimage, so g is surjective.

For the extension, if $g(x_1, y_1) = g(x_2, y_2)$, then $x_1 + y_1 = x_2 + y_2$ and $x_1 - y_1 = x_2 - y_2$. Adding and subtracting gives $x_1 = x_2$ and $y_1 = y_2$. Hence g is injective. Its inverse is

$$h\colon \mathbb{R}^2 \to \mathbb{R}^2, \qquad h(u, v) = \left(\frac{u+v}{2}, \frac{u-v}{2}\right).$$

We checked $g(h(u, v)) = (u, v)$ above. In the other order,

$$h(g(x, y)) = \left(\frac{(x+y)+(x-y)}{2}, \frac{(x+y)-(x-y)}{2}\right) = (x, y).$$

3. Bijections and inverses

Let $f\colon X \to Y$. An inverse for f is a function $h\colon Y \to X$ satisfying $h(f(x)) = x$ for all $x \in X$ and $f(h(y)) = y$ for all $y \in Y$. Prove that f is bijective if and only if it has an inverse.

(a) Assume an inverse h exists. Prove that f is injective.

(b) Under the same assumption, prove that f is surjective.

(c) Conversely, assume that f is bijective. Explain why every $y \in Y$ has exactly one preimage. Use this to define h and verify both identities.

Keep track: Which identity is used in (a), and which in (b)?

Solution. **(a)** Suppose $f(x_1) = f(x_2)$. Applying h to both sides gives $h(f(x_1)) = h(f(x_2))$, hence $x_1 = x_2$. Therefore f is injective. This uses $h \circ f = \mathrm{id}_X$.

(b) For any $y \in Y$, take $x = h(y) \in X$. Then $f(x) = f(h(y)) = y$, proving surjectivity. This uses $f \circ h = \mathrm{id}_Y$.

(c) Suppose f is bijective. For every $y \in Y$, surjectivity provides at least one $x \in X$ with $f(x) = y$, and injectivity implies that there is at most one such x. Define $h(y)$ to be this unique x. This specifies a function $h\colon Y \to X$. By definition, $f(h(y)) = y$ for every $y \in Y$. For each $x \in X$, x is the unique preimage of $f(x)$, so $h(f(x)) = x$. Thus h is an inverse. Together, (a)–(c) prove the equivalence.

4. A bijection of the plane

Let $F\colon \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $F(x,y) = (x+y, x+2y)$. Show that F is bijective by finding an inverse. State the domain and codomain of F^{-1} and check $F^{-1} \circ F$ and $F \circ F^{-1}$.

Solution. For a target (u,v), the equations $u = x+y$ and $v = x+2y$ give $y = v-u$ and $x = 2u-v$. Define

$$K\colon \mathbb{R}^2 \to \mathbb{R}^2, \qquad K(u,v) = (2u-v, v-u).$$

This is well-defined because both coordinates are real. We have

$$F(K(u,v)) = ((2u-v)+(v-u), (2u-v)+2(v-u)) = (u,v),$$
$$K(F(x,y)) = (2(x+y)-(x+2y), (x+2y)-(x+y)) = (x,y).$$

Therefore F is bijective and $F^{-1} = K$.

5. An inverse with excluded values

Consider

$$r\colon \mathbb{R}\setminus\{3\} \longrightarrow \mathbb{R}\setminus\{1\}, \qquad r(x) = \frac{x+1}{x-3}.$$

(a) Check that this formula defines a function with the stated codomain.

(b) Find a candidate inverse and state its domain and codomain.

(c) Check that the candidate has values in its claimed codomain and verify both compositions. Conclude that r is bijective.

Solution. (a) For $x \neq 3$, the denominator is nonzero. Moreover, $r(x) = 1$ would imply $x+1 = x-3$, hence $1 = -3$, a contradiction. Thus $r(x) \in \mathbb{R}\setminus\{1\}$.

(b) Solving $y = (x+1)/(x-3)$ gives $yx - 3y = x+1$, then $x(y-1) = 3y+1$. The candidate is

$$s\colon \mathbb{R}\setminus\{1\} \to \mathbb{R}\setminus\{3\}, \qquad s(y) = \frac{3y+1}{y-1}.$$

(c) For $y \neq 1$, the formula is defined, and

$$s(y) - 3 = \frac{4}{y-1} \neq 0.$$

Thus its values lie in the required codomain. Since $s(y)+1 = 4y/(y-1)$, we get $r(s(y)) = y$. Conversely, for $x \neq 3$,

$$3r(x)+1 = \frac{4x}{x-3}, \qquad r(x)-1 = \frac{4}{x-3},$$

so $s(r(x)) = x$. Therefore r is bijective and $r^{-1} = s$.

6. Intervals and cardinality

Two sets have the same cardinality if there is a bijection between them. Let $a, b \in \mathbb{R}$ with $a < b$.

(a) Prove that $L\colon (0,1) \to (a,b)$, $L(t) = a + (b-a)t$, is bijective by finding and checking its inverse. Check the interval restrictions.

(b) Find a bijection $H\colon (0,\infty) \to (0,1)$ and verify your answer. *Hint:* Try $H(x) = x/(1+x)$.

(c) Use (a) and (b) to construct a bijection from $(0,\infty)$ to (a,b). Give its inverse and state the resulting equality of cardinalities.

Solution. (a) Since $b - a > 0$, $0 < t < 1$ implies $a < a + (b-a)t < b$. For $y \in (a,b)$, set

$$L^{-1}(y) = \frac{y-a}{b-a}.$$

The inequalities $0 < y - a < b - a$ imply $L^{-1}(y) \in (0,1)$. Also $L^{-1}(L(t)) = t$ and $L(L^{-1}(y)) = y$ by substitution. Thus L is bijective.

(b) For $x > 0$, $0 < x < 1 + x$, so $H(x) = x/(1+x)$ lies in $(0,1)$. Define $J\colon (0,1) \to (0,\infty)$ by $J(t) = t/(1-t)$. Its denominator and numerator are positive. We check

$$H(J(t)) = \frac{t/(1-t)}{1 + t/(1-t)} = t, \qquad J(H(x)) = \frac{x/(1+x)}{1 - x/(1+x)} = x.$$

Hence H is bijective with inverse J.

(c) The desired function is

$$B = L \circ H\colon (0,\infty) \to (a,b), \qquad B(x) = a + (b-a)\frac{x}{1+x}.$$

Its inverse is

$$C = J \circ L^{-1}\colon (a,b) \to (0,\infty), \qquad C(y) = \frac{y-a}{b-y}.$$

Indeed $y - a > 0$ and $b - y > 0$. The two inverse identities from (a) and (b) give $C \circ B = J \circ L^{-1} \circ L \circ H = \mathrm{id}_{(0,\infty)}$ and $B \circ C = L \circ H \circ J \circ L^{-1} = \mathrm{id}_{(a,b)}$. Therefore

$$|(0,\infty)| = |(0,1)| = |(a,b)|.$$

7. Composition of bijections

Suppose $f\colon X \to Y$ and $g\colon Y \to Z$ are bijective.

(a) Prove directly that $g \circ f\colon X \to Z$ is injective.

(b) Prove directly that $g \circ f$ is surjective.

(c) Prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ by checking both compositions. Explain why the order of the inverses is reversed.

Solution. **(a)** Suppose $g(f(x_1)) = g(f(x_2))$. Injectivity of g gives $f(x_1) = f(x_2)$, and injectivity of f gives $x_1 = x_2$. Thus $g \circ f$ is injective.

(b) Let $z \in Z$. Surjectivity of g gives $y \in Y$ with $g(y) = z$. Surjectivity of f gives $x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = z$, so $g \circ f$ is surjective.

(c) Let $h = f^{-1} \circ g^{-1}\colon Z \to X$. For every $x \in X$ and $z \in Z$,

$$h((g \circ f)(x)) = f^{-1}(g^{-1}(g(f(x)))) = f^{-1}(f(x)) = x,$$
$$(g \circ f)(h(z)) = g(f(f^{-1}(g^{-1}(z)))) = g(g^{-1}(z)) = z.$$

Thus $h = (g \circ f)^{-1}$. To undo "first f, then g," we must first undo g, then undo f. The domains also require the order $Z \xrightarrow{g^{-1}} Y \xrightarrow{f^{-1}} X$.

8. a piecewise integer function

Let $p\colon \mathbb{Z} \to \mathbb{Z}$ be given by

$$p(n) = \begin{cases} n/2, & n \text{ even}, \\ (n+1)/2, & n \text{ odd}. \end{cases}$$

Is p injective? Prove your answer.

Solution. The rule defines an integer for every input: if $n = 2k$, then $p(n) = k$; if $n = 2k+1$, then $p(n) = k+1$. The parity cases are disjoint and exhaustive. It is not injective, since $1 \neq 2$ but $p(1) = 1 = p(2)$.

9. an integer-valued function

Consider $q\colon \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $q(m,n) = 3m - 4n$.

(a) Prove that q is a function.

(b) Is q injective? Prove your answer.

(c) Is q surjective? Prove your answer.

Solution. **(a)** Every ordered pair $(m,n) \in \mathbb{Z}^2$ determines exactly one integer $3m-4n$, so q is a function with the stated domain and codomain.

(b) It is not injective: $(0,0) \neq (4,3)$, but $q(0,0) = 0 = q(4,3)$.

(c) Let $z \in \mathbb{Z}$ be arbitrary. Taking $(m,n) = (-z,-z)$ gives

$$q(-z,-z) = -3z + 4z = z.$$

Both coordinates are integers, so every element of the codomain has a preimage. Therefore q is surjective.